

Homework 1

Problem 1

We filter by DHCP. We see that the client requested IP 192.168.20.11 (No. 1254), but was Not Acknowledged (No. 1255). After that the client send a Discover (No. 1256) and received an Offer (No. 1257) for the IP 192.168.30.11, then the client Requested IP 192.168.30.11 (No. 1258), which was Acknowledged (No. 1259).

dhcp						
No.	Time	Source	Destination	Protocol	Length	Info
1254	121.772905	0.0.0.0	255.255.255.255	DHCP	346	DHCP Request - Transaction ID 0x5f511e61
1255	121.782156	192.168.30.1	255.255.255.255	DHCP	346	DHCP NAK - Transaction ID 0x5f511e61
1256	121.805157	0.0.0.0	255.255.255.255	DHCP	346	DHCP Discover - Transaction ID 0x96a1041e
1257	121.810908	192.168.30.1	192.168.30.11	DHCP	361	DHCP Offer - Transaction ID 0x96a1041e
1258	121.821659	0.0.0.0	255.255.255.255	DHCP	346	DHCP Request - Transaction ID 0x96a1041e
1259	121.827910	192.168.30.1	192.168.30.11	DHCP	361	DHCP ACK - Transaction ID 0x96a1041e

The answer is: 192.168.30.11.

Problem 2

We filter by NTP and see that the source (2003:41:6012:121::10) send a NTP packet (No. 2918) and then received one from the destination (2003:41:6012:110::dcf7:123), which had the mode flag set to Server.

ntp						
No.	Time	Source	Destination	Protocol	Length	Info
157	13.806387	192.168.121.40	212.224.120.164	NTP	94	NTP Version 3, client
158	13.808394	212.224.120.164	192.168.121.40	NTP	94	NTP Version 3, server
187	15.802944	192.168.121.40	78.46.107.140	NTP	94	NTP Version 3, client
188	15.809945	78.46.107.140	192.168.121.40	NTP	94	NTP Version 3, server
459	36.808209	192.168.121.40	148.251.154.36	NTP	94	NTP Version 3, client
460	36.814964	148.251.154.36	192.168.121.40	NTP	94	NTP Version 3, server
2918	286.367467	2003:51:6012:121::10	2003:51:6012:110::dcf7:123	NTP	114	NTP Version 4, client
2919	286.368969	2003:51:6012:110::dcf7:123	2003:51:6012:121::10	NTP	114	NTP Version 4, server
3891	334.812247	192.168.121.40	212.227.54.68	NTP	94	NTP Version 3, client
3893	334.817994	212.227.54.68	192.168.121.40	NTP	94	NTP Version 3, server

The answer is: 2003:41:6012:110::dcf7:123.

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Problem 3

We filter by DNS and see that in the packet No. 2620, the authoritative name server for the domain blog.webergetz.net were ns1.hans.hosteurope.de and ns2.hans.hosteurope.de.

No.	Time	Source	Destination	Protocol	Length	Info
242	21.049259	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xb4ca A blog.webernetz.net
243	21.050522	192.168.120.22	192.168.121.2	DNS	152	Standard query response 0xb4ca A blog.webernetz.net A 5.35.226.136 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de
851	81.049328	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
913	84.047794	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
939	87.047761	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
966	90.047724	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
1427	141.050146	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
1486	144.048490	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
1514	147.048589	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
1544	150.048552	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
2023	201.051215	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xd306 A blog.webernetz.net
2091	204.049182	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xd306 A blog.webernetz.net
2118	207.049168	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xd306 A blog.webernetz.net
2154	210.050113	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xd306 A blog.webernetz.net
2619	261.052035	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x2aa5 A blog.webernetz.net
2620	261.053287	192.168.120.22	192.168.121.2	DNS	152	Standard query response 0x2aa5 A blog.webernetz.net A 5.35.226.136 NS ns1.hans.hosteurope.de NS ns2.hans.hosteurope.de
3506	321.053056	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xe597 A blog.webernetz.net
3507	321.054057	192.168.120.22	192.168.121.2	DNS	152	Standard query response 0xe597 A blog.webernetz.net A 5.35.226.136 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de
3636	322.493083	2003:51:6012:121::2	2003:51:6012:120::a08:53	DNS	100	Standard query 0xde4e AAAA ip.webernetz.net
3637	322.494205	2003:51:6012:120::a08:53	2003:51:6012:121::2	DNS	182	Standard query response 0xde4e AAAA ip.webernetz.net AAAA 2003:51:6012:110::19 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de

```
> Frame 2620: 152 bytes on wire (1216 bits), 152 bytes captured (1216 bits) on interface unknown, id 0
> Ethernet II, Src: Cisco_9e:11:41 (00:14:09:9e:11:41), Dst: Cisco_79:3f:11 (00:1e:7a:79:3f:11)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 121
> Internet Protocol Version 4, Src: 192.168.120.22, Dst: 192.168.121.2
> User Datagram Protocol, Src Port: 53, Dst Port: 56469
  Domain Name System (response)
    Transaction ID: 0x2aa5
    Flags: 0x8100 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 2
    Additional RRs: 0
    Queries
    Answers
    Authoritative nameservers
      webernetz.net: type NS, class IN, ns ns1.hans.hosteurope.de
      webernetz.net: type NS, class IN, ns ns2.hans.hosteurope.de
    [Request In: 2619]
    [Time: 0.001252000 seconds]
```

The answer is: ns1.hans.hosteurope.de.

Problem 4

We filter by CDP and see in packet No. 133 (the first packet from CCNP-LAB-S2) that the Port ID for CCNP-LAB-S2 is GigabitEthernet0/2.

No.	Time	Source	Destination	Protocol	Length	Info
133	11.080970	Cisco_a1:5a:9a	CDP/VTP/DT/PaGp/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
138	11.748317	Cisco_ae:31:99	CDP/VTP/DT/PaGp/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
761	71.098782	Cisco_a1:5a:9a	CDP/VTP/DT/PaGp/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
771	71.752519	Cisco_ae:31:99	CDP/VTP/DT/PaGp/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
1335	131.112111	Cisco_a1:5a:9a	CDP/VTP/DT/PaGp/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
1346	131.756713	Cisco_ae:31:99	CDP/VTP/DT/PaGp/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
1921	191.121684	Cisco_a1:5a:9a	CDP/VTP/DT/PaGp/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
1935	191.761025	Cisco_ae:31:99	CDP/VTP/DT/PaGp/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
2511	251.133254	Cisco_a1:5a:9a	CDP/VTP/DT/PaGp/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
2521	251.788355	Cisco_ae:31:99	CDP/VTP/DT/PaGp/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
3032	311.149194	Cisco_a1:5a:9a	CDP/VTP/DT/PaGp/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
3044	311.786290	Cisco_ae:31:99	CDP/VTP/DT/PaGp/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1

```
> Frame 133: 498 bytes on wire (3984 bits), 498 bytes captured (3984 bits) on interface unknown, id 0
> Ethernet II, Src: Cisco_a1:5a:9a (00:0a:8a:a1:5a:9a), Dst: CDP/VTP/DT/PaGp/UDLD (01:00:0c:cccccccc)
> 802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 1
> Logical-Link Control
  Cisco Discovery Protocol
    Version: 2
    TTL: 180 seconds
    Checksum: 0xde38 [correct]
    [Checksum Status: Good]
    Device ID: CCNP-LAB-S2.webernetz.net
    Addresses
    Port ID: GigabitEthernet0/2
      Type: Port ID (0x0003)
      Length: 22
      Sent through Interface: GigabitEthernet0/2
```

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The answer is: GigabitEthernet0/2.

Problem 5

We filter by CDP and see in packet No. 133 (the first packet from CCNP-LAB-S2) that the IOS Software Version for CCNP-LAB-S2 is: Version 12.1(22)EA14.

cdp							
No.	Time	Source	Destination	Protocol	Length	Info	
133	11.088970	Cisco_a1:5a:9a	CDP/VTP/DTP/PagP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net	Port ID: GigabitEthernet0/2
138	11.748317	Cisco_ae:31:99	CDP/VTP/DTP/PagP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net	Port ID: GigabitEthernet0/1
761	71.098702	Cisco_a1:5a:9a	CDP/VTP/DTP/PagP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net	Port ID: GigabitEthernet0/2
771	71.752519	Cisco_ae:31:99	CDP/VTP/DTP/PagP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net	Port ID: GigabitEthernet0/1
1335	131.112111	Cisco_a1:5a:9a	CDP/VTP/DTP/PagP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net	Port ID: GigabitEthernet0/2
1346	131.756713	Cisco_ae:31:99	CDP/VTP/DTP/PagP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net	Port ID: GigabitEthernet0/1
1921	191.121684	Cisco_a1:5a:9a	CDP/VTP/DTP/PagP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net	Port ID: GigabitEthernet0/2
1935	191.761025	Cisco_ae:31:99	CDP/VTP/DTP/PagP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net	Port ID: GigabitEthernet0/1
2511	251.133254	Cisco_a1:5a:9a	CDP/VTP/DTP/PagP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net	Port ID: GigabitEthernet0/2
2521	251.788355	Cisco_ae:31:99	CDP/VTP/DTP/PagP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net	Port ID: GigabitEthernet0/1
3282	311.149194	Cisco_a1:5a:9a	CDP/VTP/DTP/PagP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net	Port ID: GigabitEthernet0/2
3304	311.786290	Cisco_ae:31:99	CDP/VTP/DTP/PagP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net	Port ID: GigabitEthernet0/1

```
Version: 2
TTL: 180 seconds
Checksum: 0xde38 [correct]
[Checksum Status: Good]
> Device ID: CCNP-LAB-S2.webernetz.net
> Addresses
> Port ID: GigabitEthernet0/2
> Capabilities
▼ Software Version
  Type: Software version (0x0005)
  Length: 276
  Software version: Cisco Internetwork Operating System Software
  Software version: IOS (tm) C2950 Software (C2950-I6K2L2Q4-M), Version 12.1(22)EA14, RELEASE SOFTWARE (fc1)
  Software version: Technical Support: http://www.cisco.com/techsupport
  Software version: Copyright (c) 1986-2010 by cisco Systems, Inc.
  Software version: Compiled Tue 26-Oct-10 10:35 by nburra
```

The answer is: Version 12.1(22)EA14.

Problem 6

I searched online “NVRAM config” and found out this:

- Network file systems (TFTP, rcp, and FTP)

So I wanted to search by TFTP and see which contain NVRAM, so I filtered by TFTP contains "NVRAM" and found this in packet No. 3770:

!

! Last configuration change at 20:55:45 UTC Fri Mar 3 2017 by weberjoh

! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by weberjoh

! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by weberjoh

version 15.1

service timestamps debug datetime msec

service timestamps log datetime msec

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service password-encryption

!

hostname CCNP-LAB-R2

!

boot-start-marker

boot-end-marker

!

!

enable secret 5 \$1\$Z.9j\$Nvobsx9NvJzqtRLQqR.9b0

!

aaa new-model

!

!

aaa group server radius foobar

server name blubb

No.	Time	Source	Destination	Protocol	Length	Info
3778	07:07:14.1	192.168.121.2	192.168.110.10	TFTP	562	Data Packet, Block: 1

```
> Frame 3778: 562 bytes on wire (4496 bits), 562 bytes captured (4496 bits) on Interface unknown, Id 0
> Ethernet II, Src: Cisco_79:3f:11 (08:00:79:3f:11), Dst: Cisco_9e:11:42 (08:00:0e:11:42)
> 802.1Q Virtual LAN, PVID 0, DEI 0, DDI 121
> Internet Protocol Version 4, Src: 192.168.121.2, Dst: 192.168.110.10
> User Datagram Protocol, Src Port: 56695, Dst Port: 1555
> Trivial File Transfer Protocol
> Data (512 bytes)
```

0000	00 14 05 1e 11 41 00 14	7e 79 3f 11 51 00 00 79	E . A . zy? . y
0010	00 00 45 00 02 20 00 01	00 00 ff 11 51 6e c0 a0	E . . . Qn
0020	79 02 c0 a0 02 00 00 00	00 14 02 00 20 00 01	y n
0030	02 01 00 00 00 00 00 00	00 14 02 00 20 00 01	y n
0040	04 67 72 62 61 7a 69 6f	6e 3d 63 65 61 6e 67 65	l g r a t i o n ' s c h a n g e
0050	0a 13 1a 20 52 39 3e 3e	70 3a 35 2a 55 5a 63	at 2013 Sun Oct
0060	29 46 72 62 38 a0 61 72	26 33 29 32 26 31 37 a0	T r i P a r ' s 2012
0070	03 79 28 27 65 62 65 72	4a 6f 68 69 21 28 4e 6e	My website Jan 1 20
0080	31 41 6d 26 62 67 6a 6e	68 67 28 61 61 71 7a 69	Watt conf id last
0090	02 78 64 63 2a 65 6a 6e	61 7a 28 33 31 3a 30 31	Updated at 23:00
00a0	14 33 58 78 55 54 63 26	46 72 65 28 48 61 72 28	16 UTC Fri Mar
00b0	03 28 32 40 55 57 28 4e	40 28 77 61 62 65 72 6a	0 2013 y y server
00c0	0f 68 6a 21 28 4a 56 51	61 6d 28 61 6f 6a 6a 69	ah! 800 AM confi
00d0	08 28 51 22 7a 29 70 7a	6a 6a 61 6a 6a 6a 6a 6a	at last i posted a
00e0	24 28 32 71 3a 3e 32 3a	31 35 2a 55 5a 61 2a 6a	E 21:02 16 UTC F
00f0	08 28 48 62 72 20 20 20	22 2a 31 37 20 42 7a	E Mar 0 2013 M
0100	28 77 63 62 65 72 6a 6f	68 6a 70 65 72 71 69 6f	WattConf History
0110	0a 13 22 31 6a 6e 6e 6e	68 72 7a 69 62 65 6a 6e	Site's history
0120	09 6d 63 73 74 61 6a 70	73 2a 6a 65 62 75 67 6b	WattConf's history
0130	11 61 62 70 69 6a 6e 6e	6f 71 69 6a 6e 67 6b	WattConf's history
0140	22 7e 6a 63 65 2a 7a 69	6a 65 73 7a 61 6a 70	WattConf's history
0150	04 67 67 28 6a 61 7a 65	7a 69 6a 65 2a 67 75	log dat time 6a
0160	01 6a 13 65 71 70 6a 61	65 2a 70 61 71 71 71	at WattConf
0170	77 6a 2a 62 6a 63 72 70	7a 69 6a 6f 6a 6a 6a	WattConf's history
0180	0a 6f 73 7a 6a 61 6a 6e	6a 2a 61 61 6a 6a 6a	WattConf's history
0190	41 61 22 32 31 6a 21 6a	62 6f 6f 7a 2a 71 7a 61	WattConf's history
01a0	72 7a 2a 6a 62 72 6a 6e	72 6a 62 6f 7a 2a 6a	WattConf's history
01b0	0a 6a 6a 62 72 6a 6e	72 6a 62 6f 7a 2a 6a	WattConf's history

The answer is: 21:02:36 UTC Fri Mar 3 2017.